DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**CARE OF NEWBORN WITH TRANSIENT TACHYPNEA (TTN)**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-013 (v2)**		
Last Version Approval Date	**18/2/2021**	Number of Pages	5

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19 / 3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

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1.0 PURPOSE

1.1. To provide guidelines to physician on how to manage sick newborn with transient tachypnea.

1.2. To emphasize the nursing care to be rendered to patients with TTNB

2.0 DEFINITION/ABBREVIATION

2.1. **TTNB (Transient Tachypnea of the Newborn)**- is a self-limited disease of full-term or near term infants that is characterized by fast breathing that usually resolves over 24-72 hours.

2.2. **Tachypnea**- rapid breathing between 60-120 breaths per minute.

2.3. **Retraction**- is a sharp pulling in of the chest muscles during breathing.

2.4. **PROM**-Premature Rupture of the Membrane

2.5. **OGT**- Oro-Gastric Tube feeding

2.6. **VBG**-Venous Blood Gases

2.7. **ECG**-Electro Cardiogram

2.8. **PPE**- Personal Protective Equipment

2.9. **NICU**-Neonatal Intensive Care Unit

3.0 RESPONSIBILITY

3.1. **Physicians (Residents, Specialists, Consultant)** are responsible to assess and evaluate patients for transient tachypnea.

3.1.1. The physician should provide a written order of the procedure/management of TTNB to be done to the patient in the medical file.

3.2. **Nurse** (Head nurse, Charge nurse, Staff nurse) will assess and monitor respiratory rate and other vital signs of the patient.

3.2.1. The nurse will be responsible in documenting the given procedure in the patient's medical file.

4.0 POLICY

4.1. All newborn patients who develop non-infectious acute respiratory disease must be carefully placed under observation in Nursery and will be subjected for admission to Neonatal Intensive Care Unit if indicated according to the newborn's clinical condition and physician's written order.

4.2. The physician should review the antenatal, delivery and postnatal history of the newborn's mother including key factors such as:

4.2.1. PROM

4.2.2. Caesarean delivery or rapid vaginal delivery

4.2.3. Maternal pyrexia

4.2.4. Maternal asthma and smoking

4.2.5. Maternal sedation

4.2.6. History of feeds

4.2.7. Known anomalies

4.3. Careful assessment must be given to infants with TTNB that comes out within the first few hours of life which present the following:

4.3.1. **CLINICALPRESENTATION**

4.3.1.1. Manifests in full Term infants, Large for gestational age babies

4.3.1.2. Tachypnea (60-120 per minute)

4.3.1.3. Presence of nasal flaring, grunting, retractions

4.3.1.4. Barrel chest (symmetric hyperinflation) for some infants

4.3.1.5. Cyanosis which appears to resolve with less than 40%0 oxygen

4.4. Blood investigations such as full blood count, septic screening and blood gases must be done.

4.5. Imaging studies such as chest X-ray, ECG and Echocardiography (if suspecting Congenital heart disease) must be ordered by the physician.

4.6. Supplemental Oxygen is given depending on the patient's ability to maintain adequate arterial oxygen concentration.

4.7. Empiric antibiotics should be started until infection has been ruled out. Intravenous ampicillin and gentamycin are used.

4.8. Gavage feedings are given until the respiratory rate has decreased enough to allow oral feedings. OGT feedings should be on hold if respiratory rate is above 80 breaths per minute.

4.9. Intravenous fluids should be administered while on restriction of feedings.

4.10. Nursing care must be carefully rendered depending on the physician's written order and management of TTNB such as:

4.10.1. Continuous assessment and monitoring of the patient's respiratory rate and other vital signs.

4.10.2. Maintaining a neutral thermal environment with minimal stimulation to the sick newborn.

4.10.3. Intravenous fluids must be properly checked to avoid fluid overload that can worsen respiratory manifestations.

5.0 PROCEDURE (S)

5.1. The Physician and nurse will assess patient for transient tachypnea of the newborn.

5.2. The physician will provide the written order of the procedure to be done to manage or treat TTNB.

5.3. The nurse will verify and carry out the doctor's order.

5.4. Identify the patient correctly.

5.5. Do hand hygiene and wear appropriate PPE according to the procedure to be done

5.6. The nurse will proceed in performing the procedure to manage TTNB according to physician's written order and policy.

5.6.1. Observation and monitoring of Vital sign

5.6.2. Oxygen Therapy

5.6.3. Pulse Oximetry monitoring

5.6.4. Suctioning

5.6.5. Antibiotic Therapy

5.7. Perform continuous re-assessment and follow -up Blood culture and CRP report of the sick newborn.

5.8. The physician will change the line of antibiotics, if blood culture and CRP are positive.

5.9. The physician and nurse will continue to re-evaluate the newborn's condition.

5.10. If the newborn remains vitally and clinically stable, stop antibiotic therapy if blood culture and CRP reports are negative, according to physician's written order.

5.11. The nurse will provide the appropriate nursing care for the newborn patient according to physician's order and management.

5.12. Document all the procedures done in the patient's medical file

5.13. Careful assessments must be given to infants with TTNB that comes out within the first few hours of life which present the following decreased enough to allow oral feedings. OGT feedings should be on hold if respiratory rate is above 80 breaths per minute.

6.0 EQUIPMENTS

6.1. PPE
6.2. Resuscitation and suction equipment

6.3. Vital signs machine
6.4. Oxygen cylinder / oxygen mask

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.2. Physician Order Form (GDOH-INP-POF-079)
7.3. Patients Medical File

8.0 REFERENCES

8.1. Newborn Services Guidelines and Protocols in the Royal College of Pediatric Website

8.2. National Institute for Health & Clinical Excellence UK

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) NICU standards, 3rd version, January 2016

8.4. DPP-NICU-0012(v2) - Care of Newborn with Transient Tachypnea, PMSH,21/10/1441

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

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Al-Mikhwah General Hospital

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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Gender: ☐ Male ☐ Female الجنس:

PHYSICIAN ORDER FORM

ALLERGIC TO:

DATE	TIME	PHYSICIAN ORDERS	PHYSICIAN STAMP / SIGNATURE	RN INITIAL / JOB NUMBER